

Package ‘rdrobust’

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Title Robust Data-Driven Statistical Inference in
Regression-Discontinuity Designs

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Description Regression-discontinuity (RD) designs are quasi-experimental research designs popular in social, behavioral and natural sciences. The RD design is usually employed to study the (local) causal effect of a treatment, intervention or policy. This package provides tools for data-driven graphical and analytical statistical inference in RD designs: `rdrobust()` to construct local-polynomial point estimators and robust confidence intervals for average treatment effects at the cutoff in Sharp, Fuzzy and Kink RD settings, `rdbwselect()` to perform bandwidth selection for the different procedures implemented, and `rdplot()` to conduct exploratory data analysis (RD plots).

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Author Sebastian Calonico [aut, cre],
Matias D. Cattaneo [aut],
Max H. Farrell [aut]

Maintainer Sebastian Calonico <scalonico@ucdavis.edu>

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rdrobust-package*Robust Data-Driven Statistical Inference in RD Designs*

Description

Regression-discontinuity (RD) designs are quasi-experimental research designs popular in social, behavioral and natural sciences. The RD design is usually employed to study the (local) causal effect of a treatment, intervention or policy. This package provides tools for data-driven graphical and analytical statistical inference in RD designs: [rdrobust](#) to construct local-polynomial point estimators and robust confidence intervals for average treatment effects at the cutoff in Sharp, Fuzzy and Kink RD settings, [rdbwselect](#) to perform bandwidth selection for the different procedures implemented, and [rdplot](#) to conduct exploratory data analysis (RD plots).

Details

Package: rdrobust
Type: Package
Version: 3.0.0
Date: 2025-07-19
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Function for statistical inference: [rdrobust](#)
Function for bandwidths selection: [rdbwselect](#)
Function for exploratory data analysis (RD plots): [rdplot](#)

Author(s)

Sebastian Calonico, University of California, Davis, CA. <scalonico@ucdavis.edu>.
Matias D. Cattaneo, Princeton University, Princeton, NJ. <cattaneo@princeton.edu>.
Max H. Farrell, University of California, Santa Barbara, CA. <maxhfarrell@ucsb.edu>.
Rocio Titiunik, Princeton University, Princeton, NJ. <titiunik@princeton.edu>.

rdbwselect	<i>Bandwidth Selection Procedures for Local Polynomial Regression Discontinuity Estimators</i>
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Description

`rdbwselect` implements bandwidth selectors for local polynomial Regression Discontinuity (RD) point estimators and inference procedures developed in Calonico, Cattaneo and Titiunik (2014a), Calonico, Cattaneo and Farrell (2018), Calonico, Cattaneo, Farrell and Titiunik (2019) and Calonico, Cattaneo and Farrell (2020).

Companion commands are: `rdrobust` for point estimation and inference procedures, and `rdplot` for data-driven RD plots (see Calonico, Cattaneo and Titiunik (2015a) for details).

A detailed introduction to this command is given in Calonico, Cattaneo and Titiunik (2015b) and Calonico, Cattaneo, Farrell and Titiunik (2019). A companion Stata package is described in Calonico, Cattaneo and Titiunik (2014b).

For more details, and related Stata and R packages useful for analysis of RD designs, visit <https://rdpackages.github.io/>

Usage

```
rdbwselect(y, x, c = NULL, fuzzy = NULL,
           deriv = NULL, p = NULL, q = NULL,
           covs = NULL, covs_drop = TRUE, ginv.tol = 1e-20,
           kernel = "tri", weights = NULL, bwselect = "mserd",
           vce = "nn", cluster = NULL, nnmatch = 3,
           scaleregul = 1, sharpbw = FALSE,
           all = NULL, subset = NULL,
           masspoints = "adjust", bwcheck = NULL,
           bwrestrict = TRUE, stdvars = FALSE)
```

Arguments

<code>y</code>	is the dependent variable.
<code>x</code>	is the running variable (a.k.a. score or forcing variable).
<code>c</code>	specifies the RD cutoff in <code>x</code> ; default is <code>c = 0</code> .
<code>fuzzy</code>	specifies the treatment status variable used to implement fuzzy RD estimation (or Fuzzy Kink RD if <code>deriv=1</code> is also specified). Default is Sharp RD design and hence this option is not used.
<code>deriv</code>	specifies the order of the derivative of the regression functions to be estimated. Default is <code>deriv=0</code> (for Sharp RD, or for Fuzzy RD if <code>fuzzy</code> is also specified). Setting <code>deriv=1</code> results in estimation of a Kink RD design (up to scale), or Fuzzy Kink RD if <code>fuzzy</code> is also specified.
<code>p</code>	specifies the order of the local-polynomial used to construct the point-estimator; default is <code>p = 1</code> (local linear regression).

<code>q</code>	specifies the order of the local-polynomial used to construct the bias-correction; default is <code>q = 2</code> (local quadratic regression).
<code>covs</code>	specifies additional covariates to be used for estimation and inference.
<code>covs_drop</code>	if TRUE, it checks for collinear additional covariates and drops them. Default is TRUE.
<code>ginv.tol</code>	tolerance used to invert matrices involving covariates when <code>covs_drop=TRUE</code> .
<code>kernel</code>	is the kernel function used to construct the local-polynomial estimator(s). Options are <code>triangular</code> (default option), <code>epanechnikov</code> and <code>uniform</code> .
<code>weights</code>	is the variable used for optional weighting of the estimation procedure. The unit-specific weights multiply the kernel function.
<code>bwselect</code>	specifies the bandwidth selection procedure to be used. Options are: <code>mserd</code> one common MSE-optimal bandwidth selector for the RD treatment effect estimator. <code>msetwo</code> two different MSE-optimal bandwidth selectors (below and above the cutoff) for the RD treatment effect estimator. <code>msesum</code> one common MSE-optimal bandwidth selector for the sum of regression estimates (as opposed to difference thereof). <code>msecomb1</code> for <code>min(mserd,msesum)</code>. <code>msecomb2</code> for <code>median(msetwo,mserd,msesum)</code>, for each side of the cutoff separately. <code>cerd</code> one common CER-optimal bandwidth selector for the RD treatment effect estimator. <code>certwo</code> two different CER-optimal bandwidth selectors (below and above the cutoff) for the RD treatment effect estimator. <code>cersum</code> one common CER-optimal bandwidth selector for the sum of regression estimates (as opposed to difference thereof). <code>cercomb1</code> for <code>min(cerd,cersum)</code>. <code>cercomb2</code> for <code>median(certwo,cerd,cersum)</code>, for each side of the cutoff separately. Note: MSE = Mean Square Error; CER = Coverage Error Rate. Default is <code>bwselect=mserd</code>. For details on implementation see Calonico, Cattaneo and Titiunik (2014a), Calonico, Cattaneo and Farrell (2018), and Calonico, Cattaneo, Farrell and Titiunik (2017), and the companion software articles.
<code>vce</code>	specifies the procedure used to compute the variance-covariance matrix estimator. Options are: <code>nn</code> for heteroskedasticity-robust nearest neighbor variance estimator with <code>nnmatch</code> the (minimum) number of neighbors to be used. <code>hc0</code> for heteroskedasticity-robust plug-in residuals variance estimator without weights. <code>hc1</code> for heteroskedasticity-robust plug-in residuals variance estimator with <code>hc1</code> weights. <code>hc2</code> for heteroskedasticity-robust plug-in residuals variance estimator with <code>hc2</code> weights.

	hc3 for heteroskedasticity-robust plug-in residuals variance estimator with hc3 weights. Default is vce=nn.
cluster	indicates the cluster ID variable used for cluster-robust variance estimation with degrees-of-freedom weights. By default it is combined with vce=nn for cluster-robust nearest neighbor variance estimation. Another option is plug-in residuals combined with vce=hc0.
nnmatch	to be combined with for vce=nn for heteroskedasticity-robust nearest neighbor variance estimator with nnmatch indicating the minimum number of neighbors to be used. Default is nnmatch=3
scaleregul	specifies scaling factor for the regularization term added to the denominator of the bandwidth selectors. Setting scaleregul = 0 removes the regularization term from the bandwidth selectors; default is scaleregul = 1.
sharpbw	option to perform fuzzy RD estimation using a bandwidth selection procedure for the sharp RD model. This option is automatically selected if there is perfect compliance at either side of the threshold.
all	if specified, rdbwselect reports all available bandwidth selection procedures.
subset	an optional vector specifying a subset of observations to be used.
masspoints	checks and controls for repeated observations in the running variable. Options are: (i) off: ignores the presence of mass points; (ii) check: looks for and reports the number of unique observations at each side of the cutoff. (iii) adjust: controls that the preliminary bandwidths used in the calculations contain a minimal number of unique observations. By default it uses 10 observations, but it can be manually adjusted with the option bwcheck). Default option is masspoints=adjust.
bwcheck	if a positive integer is provided, the preliminary bandwidth used in the calculations is enlarged so that at least bwcheck unique observations are used.
bwrestrict	if TRUE, computed bandwidths are restricted to lie within the range of x; default is bwrestrict = TRUE.
stdvars	if TRUE, x and y are standardized before computing the bandwidths; default is stdvars = FALSE.

Value

N	vector with sample sizes to the left and to the right of the cutoff.
c	cutoff value.
p	order of the local-polynomial used to construct the point-estimator.
q	order of the local-polynomial used to construct the bias-correction estimator.
bws	matrix containing the estimated bandwidths for each selected procedure.
bwselect	bandwidth selection procedure employed.
kernel	kernel function used to construct the local-polynomial estimator(s).

Author(s)

Sebastian Calonico, University of California, Davis, CA. <scalonico@ucdavis.edu>.
 Matias D. Cattaneo, Princeton University, Princeton, NJ. <cattaneo@princeton.edu>.
 Max H. Farrell, University of California, Santa Barbara, CA. <maxhfarrell@ucsb.edu>.
 Rocio Titiunik, Princeton University, Princeton, NJ. <titiunik@princeton.edu>.

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- Calonico, S., M. D. Cattaneo, and R. Titiunik. 2014b. **Robust Data-Driven Inference in the Regression-Discontinuity Design**. *Stata Journal* 14(4): 909-946.
- Calonico, S., M. D. Cattaneo, and R. Titiunik. 2015a. **Optimal Data-Driven Regression Discontinuity Plots**. *Journal of the American Statistical Association* 110(512): 1753-1769.
- Calonico, S., M. D. Cattaneo, and R. Titiunik. 2015b. **rdrobust: An R Package for Robust Nonparametric Inference in Regression-Discontinuity Designs**. *R Journal* 7(1): 38-51.
- Cattaneo, M. D., B. Frandsen, and R. Titiunik. 2015. **Randomization Inference in the Regression Discontinuity Design: An Application to the Study of Party Advantages in the U.S. Senate**. *Journal of Causal Inference* 3(1): 1-24.

See Also

[rdrobust](#), [rdplot](#)

Examples

```
x <- runif(1000,-1,1)
y <- 5 + 3*x + 2*(x>=0) + rnorm(1000)
rdbwselect(y,x)
```

rdbwselect_2014	<i>Deprecated Bandwidth Selection Procedures for Local-Polynomial Regression-Discontinuity Estimators.</i>
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Description

`rdbwselect_2014` is a deprecated command implementing three bandwidth selectors for local polynomial Regression Discontinuity (RD) point estimators and inference procedures, as described in Calonico, Cattaneo and Titiunik (2014).

This command is no longer supported or updated, and it is made available only for backward compatibility purposes. Please use `rdbwselect` instead.

The latest version of the `rdrobust` package includes the following commands: `rdrobust` for point estimation and inference procedures. `rdbwselect` for data-driven bandwidth selection. `rdplot` for data-driven RD plots.

For more details, and related Stata and R packages useful for analysis of RD designs, visit <https://rdpackages.github.io/>

Usage

```
rdbwselect_2014(y, x, subset = NULL, c = 0, p = 1, q = 2, deriv = 0,
               rho = NULL, kernel = "tri", bwselect = "CCT", scaleregul = 1,
               delta = 0.5, cvgrid_min = NULL, cvgrid_max = NULL,
               cvgrid_length = NULL, cvplot = FALSE, vce = "nn", matches = 3,
               all = FALSE, precalc = TRUE )
```

Arguments

<code>y</code>	is the dependent variable.
<code>x</code>	is the running variable (a.k.a. score or forcing variable).
<code>subset</code>	an optional vector specifying a subset of observations to be used.
<code>c</code>	specifies the RD cutoff in <code>x</code> ; default is <code>c = 0</code> .
<code>p</code>	specifies the order of the local-polynomial used to construct the point-estimator; default is <code>p = 1</code> (local linear regression).
<code>q</code>	specifies the order of the local-polynomial used to construct the bias-correction; default is <code>q = 2</code> (local quadratic regression).
<code>deriv</code>	specifies the order of the derivative of the regression function to be estimated; default is <code>deriv = 0</code> (Sharp RD, or Fuzzy RD if <code>fuzzy</code> is also specified). Setting it equal to 1 results in estimation of a Kink RD design (or Fuzzy Kink RD if <code>fuzzy</code> is also specified).
<code>rho</code>	if specified, sets the pilot bandwidth <code>b</code> equal to <code>h/rho</code> , where <code>h</code> is computed using the method and options chosen below.
<code>kernel</code>	is the kernel function used to construct the local-polynomial estimator(s). Options are <code>triangular</code> (default option), <code>epanechnikov</code> and <code>uniform</code> .

bwselect	selects the bandwidth selection procedure to be used. By default it computes both h and b, unless ρ is specified, in which case it only computes h and sets $b = h/\rho$. Options are: CCT for bandwidth selector proposed by Calonico, Cattaneo and Titiunik (2014) (default option). IK for bandwidth selector proposed by Imbens and Kalyanaraman (2012) (only available for Sharp RD design). CV for cross-validation method proposed by Ludwig and Miller (2007) (only available for Sharp RD design).
scaleregul	specifies scaling factor for the regularization terms of CCT and IK bandwidth selectors. Setting <code>scaleregul = 0</code> removes the regularization term from the bandwidth selectors; default is <code>scaleregul = 1</code> .
delta	sets the quantile that defines the sample used in the cross-validation procedure. This option is used only if <code>bwselect(CV)</code> is specified; default is <code>delta = 0.5</code> , that is, the median of the control and treated samples.
cvgrid_min	sets the minimum value of the bandwidth grid used in the cross-validation procedure. This option is used only if <code>bwselect = "CV"</code> is specified.
cvgrid_max	sets the maximum value of the bandwidth grid used in the cross-validation procedure. This option is used only if <code>bwselect = "CV"</code> is specified.
cvgrid_length	sets the bin length of the (evenly-spaced) bandwidth grid used in the cross-validation procedure. This option is used only if <code>bwselect = "CV"</code> is specified.
cvplot	generates a graph of the CV objective function. This option is used only if <code>bwselect = "CV"</code> is specified.
vce	specifies the procedure used to compute the variance-covariance matrix estimator. This option is used only if CCT or IK bandwidth procedures are employed. Options are: <code>nn</code> for nearest-neighbor matches residuals using matches number of matches. This is the default option (with <code>matches = 3</code>, see below). <code>resid</code> for estimated plug-in residuals using h bandwidth.
matches	specifies the number of matches in the nearest-neighbor based variance-covariance matrix estimator. This option is used only when nearest-neighbor matches residuals are employed; default is <code>matches = 3</code> .
all	if specified, <code>rdbwselect_2014</code> reports three different procedures: CCT for bandwidth selector proposed by Calonico, Cattaneo and Titiunik (2014). IK for bandwidth selector proposed by Imbens and Kalyanaraman (2012). CV for cross-validation method proposed by Ludwig and Miller (2007).
precalc	internal option.

Value

bws	matrix containing the estimated bandwidths for each selected procedure.
bwselect	bandwidth selection procedure employed.
kernel	kernel function used to construct the local-polynomial estimator(s).
p	order of the local-polynomial used to construct the point-estimator.
q	order of the local-polynomial used to construct the bias-correction estimator.

Author(s)

Sebastian Calonico, University of California, Davis, CA. <scalonico@ucdavis.edu>.

Matias D. Cattaneo, Princeton University, Princeton, NJ. <cattaneo@princeton.edu>.

Max H. Farrell, University of California, Santa Barbara, CA. <maxhfarrell@ucsb.edu>.

Rocio Titiunik, Princeton University, Princeton, NJ. <titiunik@princeton.edu>.

References

Calonico, S., Cattaneo, M. D., and R. Titiunik. 2014. **Robust Nonparametric Confidence Intervals for Regression-Discontinuity Designs**. *Econometrica* 82(6): 2295-2326. .

See Also

[rdrobust](#), [rdplot](#)

Examples

```
x<-runif(1000,-1,1)
y<-5+3*x+2*(x>=0)+rnorm(1000)
rdbwselect_2014(y,x)
```

rdplot

Data-Driven Regression Discontinuity Plots

Description

[rdplot](#) implements several data-driven Regression Discontinuity (RD) plots, using either evenly-spaced or quantile-spaced partitioning. Two type of RD plots are constructed: (i) RD plots with binned sample means tracing out the underlying regression function, and (ii) RD plots with binned sample means mimicking the underlying variability of the data. For technical and methodological details see Calonico, Cattaneo and Titiunik (2015a).

Companion commands are: [rdrobust](#) for point estimation and inference procedures, and [rdbwselect](#) for data-driven bandwidth selection.

A detailed introduction to this command is given in Calonico, Cattaneo and Titiunik (2015b) and Calonico, Cattaneo, Farrell and Titiunik (2017). A companion Stata package is described in Calonico, Cattaneo and Titiunik (2014).

For more details, and related Stata and R packages useful for analysis of RD designs, visit <https://rdpackages.github.io/>

Usage

```
rdplot(y, x, c = 0, p = 4, nbins = NULL, binselect = "esmv",
       scale = NULL, kernel = "uni", weights = NULL, h = NULL,
       covs = NULL, covs_eval = "mean", covs_drop = TRUE, ginv.tol = 1e-20,
       support = NULL, subset = NULL, masspoints = "adjust",
       hide = FALSE, ci = NULL, shade = FALSE, title = NULL,
       x.label = NULL, y.label = NULL, x.lim = NULL, y.lim = NULL,
       col.dots = NULL, col.lines = NULL)
```

Arguments

y	is the dependent variable.
x	is the running variable (a.k.a. score or forcing variable).
c	specifies the RD cutoff in x; default is $c = 0$.
p	specifies the order of the global-polynomial used to approximate the population conditional mean functions for control and treated units; default is $p = 4$.
nbins	specifies the number of bins used to the left of the cutoff, denoted J_- , and to the right of the cutoff, denoted J_+ , respectively. If not specified, J_+ and J_- are estimated using the method and options chosen below.
binselect	specifies the procedure to select the number of bins. This option is available only if J_- and J_+ are not set manually. Options are: es: IMSE-optimal evenly-spaced method using spacings estimators. espr: IMSE-optimal evenly-spaced method using polynomial regression. esmv: mimicking variance evenly-spaced method using spacings estimators. This is the default option. esmvpr: mimicking variance evenly-spaced method using polynomial regression. qs: IMSE-optimal quantile-spaced method using spacings estimators. qspr: IMSE-optimal quantile-spaced method using polynomial regression. qsmv: mimicking variance quantile-spaced method using spacings estimators. qsmvpr: mimicking variance quantile-spaced method using polynomial regression.
scale	specifies a multiplicative factor to be used with the optimal numbers of bins selected. Specifically, the number of bins used for the treatment and control groups will be $scale \times \hat{J}_+$ and $scale \times \hat{J}_-$, where $\hat{J}$ denotes the estimated optimal numbers of bins originally computed for each group; default is $scale = 1$.
kernel	specifies the kernel function used to construct the local-polynomial estimator(s). Options are: triangular, epanechnikov, and uniform. Default is kernel=uniform (i.e., equal/no weighting to all observations on the support of the kernel).
weights	is the variable used for optional weighting of the estimation procedure. The unit-specific weights multiply the kernel function.
h	specifies the bandwidth used to construct the (global) polynomial fits given the kernel choice kernel. If not specified, the bandwidths are chosen to span the

	full support of the data. If two bandwidths are specified, the first bandwidth is used for the data below the cutoff and the second bandwidth is used for the data above the cutoff.
covs	specifies additional covariates to be used in the polynomial regression.
covs_eval	sets the evaluation points for the additional covariates, when included in the estimation. Options are: covs_eval = 0 and covs_eval = "mean" (default)
covs_drop	if TRUE, it checks for collinear additional covariates and drops them. Default is TRUE.
ginv.tol	tolerance used to invert matrices involving covariates when covs_drop=TRUE.
support	specifies an optional extended support of the running variable to be used in the construction of the bins; default is the sample range.
subset	an optional vector specifying a subset of observations to be used.
masspoints	checks and controls for repeated observations in the running variable. Options are: (i) off: ignores the presence of mass points; (ii) check: looks for and reports the number of unique observations at each side of the cutoff. (iii) adjust: sets binselect() as polynomial regression when mass points are present. Default option is masspoints=adjust.
hide	logical. If TRUE, it omits the RD plot; default is hide = FALSE.
ci	optional graphical option to display confidence intervals of selected level for each bin.
shade	optional graphical option to replace confidence intervals with shaded areas.
title	optional title for the RD plot.
x.label	optional label for the x-axis of the RD plot.
y.label	optional label for the y-axis of the RD plot.
x.lim	optional setting for the range of the x-axis in the RD plot.
y.lim	optional setting for the range of the y-axis in the RD plot.
col.dots	optional setting for the color of the dots in the RD plot.
col.lines	optional setting for the color of the lines in the RD plot.

Value

binselect	method used to compute the optimal number of bins.
N	sample sizes used to the left and right of the cutoff.
Nh	effective sample sizes used to the left and right of the cutoff.
c	cutoff value.
p	order of the global polynomial used.
h	bandwidth used to the left and right of the cutoff.
kernel	kernel used.

J	selected number of bins to the left and right of the cutoff.
J_IMSE	IMSE optimal number of bins to the left and right of the cutoff.
J_MV	Mimicking variance number of bins to the left and right of the cutoff.
coef	matrix containing the coefficients of the p^{th} order global polynomial estimated both sides of the cutoff.
coef_covs	coefficients of the additional covariates, only returned when covs() are used.
scale	selected scale value.
rscale	implicit scale value.
bin_avg	average bin length.
bin_med	median bin length.
vars_bins	data frame containing the variables used to construct the bins: bin id, cutoff values, mean of x and y within each bin, cutoff points and confidence interval bounds.
vars_poly	data frame containing the variables used to construct the global polynomial plot.
rdplot	a standard ggplot object that can be used for further customization.

Author(s)

Sebastian Calonico, University of California, Davis, CA. <scalonico@ucdavis.edu>.
 Matias D. Cattaneo, Princeton University, Princeton, NJ. <cattaneo@princeton.edu>.
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References

Calonico, S., M. D. Cattaneo, M. H. Farrell, and R. Titiunik. 2017. **rdrobust: Software for Regression Discontinuity Designs**. *Stata Journal* 17(2): 372-404.
 Calonico, S., M. D. Cattaneo, and R. Titiunik. 2014. **Robust Data-Driven Inference in the Regression-Discontinuity Design**. *Stata Journal* 14(4): 909-946.
 Calonico, S., M. D. Cattaneo, and R. Titiunik. 2015a. **Optimal Data-Driven Regression Discontinuity Plots**. *Journal of the American Statistical Association* 110(512): 1753-1769.
 Calonico, S., M. D. Cattaneo, and R. Titiunik. 2015b. **rdrobust: An R Package for Robust Nonparametric Inference in Regression-Discontinuity Designs**. *R Journal* 7(1): 38-51.
 Cattaneo, M. D., B. Frandsen, and R. Titiunik. 2015. **Randomization Inference in the Regression Discontinuity Design: An Application to the Study of Party Advantages in the U.S. Senate**. *Journal of Causal Inference* 3(1): 1-24.

See Also

[rdbwselect](#), [rdrobust](#)

Examples

```
x <- runif(1000, -1, 1)
y <- 5 + 3*x + 2*(x>=0) + rnorm(1000)
rdplot(y, x)
```

rdrobust

*Local-Polynomial RD Estimation with Robust Confidence Intervals***Description**

`rdrobust` implements local polynomial Regression Discontinuity (RD) point estimators with robust bias-corrected confidence intervals and inference procedures developed in Calonico, Cattaneo and Titiunik (2014a), Calonico, Cattaneo and Farrell (2018), Calonico, Cattaneo, Farrell and Titiunik (2019), and Calonico, Cattaneo and Farrell (2020). It also computes alternative estimation and inference procedures available in the literature.

Companion commands are: `rdbwselect` for data-driven bandwidth selection, and `rdplot` for data-driven RD plots (see Calonico, Cattaneo and Titiunik (2015a) for details).

A detailed introduction to this command is given in Calonico, Cattaneo and Titiunik (2015b), and Calonico, Cattaneo, Farrell and Titiunik (2017). A companion Stata package is described in Calonico, Cattaneo and Titiunik (2014b).

For more details, and related Stata and R packages useful for analysis of RD designs, visit <https://rdpackages.github.io/>

Usage

```
rdrobust(y, x, c = NULL, fuzzy = NULL,
         deriv = NULL, p = NULL, q = NULL,
         h = NULL, b = NULL, rho = NULL,
         covs = NULL, covs_drop = TRUE, ginv.tol = 1e-20,
         kernel = "tri", weights = NULL, bwselect = "mserd",
         vce = "nn", cluster = NULL,
         nnmatch = 3, level = 95, scalepar = 1, scaleregul = 1,
         sharpbw = FALSE, detail = NULL, all = NULL, subset = NULL,
         masspoints = "adjust", bwcheck = NULL,
         bwrestrict = TRUE, stdvars = FALSE)
```

Arguments

<code>y</code>	is the dependent variable.
<code>x</code>	is the running variable (a.k.a. score or forcing variable).
<code>c</code>	specifies the RD cutoff in <code>x</code> ; default is <code>c = 0</code> .
<code>fuzzy</code>	specifies the treatment status variable used to implement fuzzy RD estimation (or Fuzzy Kink RD if <code>deriv=1</code> is also specified). Default is Sharp RD design and hence this option is not used.
<code>deriv</code>	specifies the order of the derivative of the regression functions to be estimated. Default is <code>deriv=0</code> (for Sharp RD, or for Fuzzy RD if <code>fuzzy</code> is also specified). Setting <code>deriv=1</code> results in estimation of a Kink RD design (up to scale), or Fuzzy Kink RD if <code>fuzzy</code> is also specified.
<code>p</code>	specifies the order of the local-polynomial used to construct the point-estimator; default is <code>p = 1</code> (local linear regression).

q	specifies the order of the local-polynomial used to construct the bias-correction; default is $q = 2$ (local quadratic regression).
h	specifies the main bandwidth used to construct the RD point estimator. If not specified, bandwidth h is computed by the companion command <code>rdbwselect</code> . If two bandwidths are specified, the first bandwidth is used for the data below the cutoff and the second bandwidth is used for the data above the cutoff.
b	specifies the bias bandwidth used to construct the bias-correction estimator. If not specified, bandwidth b is computed by the companion command <code>rdbwselect</code> . If two bandwidths are specified, the first bandwidth is used for the data below the cutoff and the second bandwidth is used for the data above the cutoff.
rho	specifies the value of ρ , so that the bias bandwidth b equals h/ρ . Default is $\rho = 1$ if h is specified but b is not.
covs	additional covariates to be used for efficiency improvements.
covs_drop	if TRUE, it checks for collinear additional covariates and drops them. Default is TRUE.
ginv.tol	tolerance used to invert matrices involving covariates when <code>covs_drop=TRUE</code> .
kernel	is the kernel function used to construct the local-polynomial estimator(s). Options are <code>triangular</code> (default option), <code>epanechnikov</code> and <code>uniform</code> .
weights	is the variable used for optional weighting of the estimation procedure. The unit-specific weights multiply the kernel function.
bwselect	specifies the bandwidth selection procedure to be used. By default it computes both h and b , unless ρ is specified, in which case it only computes h and sets $b=h/\rho$.

Options are:

`mserd` one common MSE-optimal bandwidth selector for the RD treatment effect estimator.

`msetwo` two different MSE-optimal bandwidth selectors (below and above the cutoff) for the RD treatment effect estimator.

`msesum` one common MSE-optimal bandwidth selector for the sum of regression estimates (as opposed to difference thereof).

`msecomb1` for $\min(\text{mserd}, \text{msesum})$.

`msecomb2` for $\text{median}(\text{msetwo}, \text{mserd}, \text{msesum})$, for each side of the cutoff separately.

`cerrd` one common CER-optimal bandwidth selector for the RD treatment effect estimator.

`certwo` two different CER-optimal bandwidth selectors (below and above the cutoff) for the RD treatment effect estimator.

`cersum` one common CER-optimal bandwidth selector for the sum of regression estimates (as opposed to difference thereof).

`cercomb1` for $\min(\text{cerrd}, \text{cersum})$.

`cercomb2` for $\text{median}(\text{certwo}, \text{cerrd}, \text{cersum})$, for each side of the cutoff separately.

Note: MSE = Mean Square Error; CER = Coverage Error Rate. Default is `bwselect=mserd`. For details on implementation see Calonico, Cattaneo and Titiunik (2014a), Calonico, Cattaneo and Farrell (2018), and Calonico, Cattaneo, Farrell and Titiunik (2019), and the companion software articles.

vce	specifies the procedure used to compute the variance-covariance matrix estimator. Options are: nn for heteroskedasticity-robust nearest neighbor variance estimator with nnmatch the (minimum) number of neighbors to be used. hc0 for heteroskedasticity-robust plug-in residuals variance estimator without weights. hc1 for heteroskedasticity-robust plug-in residuals variance estimator with hc1 weights. hc2 for heteroskedasticity-robust plug-in residuals variance estimator with hc2 weights. hc3 for heteroskedasticity-robust plug-in residuals variance estimator with hc3 weights. Default is vce=nn.
cluster	indicates the cluster ID variable used for cluster-robust variance estimation with degrees-of-freedom weights. By default it is combined with vce=nn for cluster-robust nearest neighbor variance estimation. Another option is plug-in residuals combined with vce=hc0.
nnmatch	to be combined with for vce=nn for heteroskedasticity-robust nearest neighbor variance estimator with nnmatch indicating the minimum number of neighbors to be used. Default is nnmatch=3
level	sets the confidence level for confidence intervals; default is level = 95.
scalepar	specifies scaling factor for RD parameter of interest. This option is useful when the population parameter of interest involves a known multiplicative factor (e.g., sharp kink RD). Default is scalepar = 1 (no scaling).
scaleregul	specifies scaling factor for the regularization term added to the denominator of the bandwidth selectors. Setting scaleregul = 0 removes the regularization term from the bandwidth selectors; default is scaleregul = 1.
sharpbw	option to perform fuzzy RD estimation using a bandwidth selection procedure for the sharp RD model. This option is automatically selected if there is perfect compliance at either side of the cutoff.
detail	if specified, rdrobust reports the output from versions 2.2.0 and earlier.
all	if specified, rdrobust reports three different procedures: (i) conventional RD estimates with conventional standard errors. (ii) bias-corrected estimates with conventional standard errors. (iii) bias-corrected estimates with robust standard errors.
subset	an optional vector specifying a subset of observations to be used.
masspoints	checks and controls for repeated observations in the running variable. Options are: (i) off: ignores the presence of mass points; (ii) check: looks for and reports the number of unique observations at each side of the cutoff. (iii) adjust: controls that the preliminary bandwidths used in the calculations contain a minimal number of unique observations. By default it uses 10 observations, but it can be manually adjusted with the option bwcheck). Default option is masspoints=adjust.

bwcheck	if a positive integer is provided, the preliminary bandwidth used in the calculations is enlarged so that at least bwcheck unique observations are used.
bwrestrict	if TRUE, computed bandwidths are restricted to lie within the range of x; default is bwrestrict = TRUE.
stdvars	if TRUE, x and y are standardized before computing the bandwidths; default is stdvars = FALSE.

Value

N	vector with the sample sizes used to the left and to the right of the cutoff.
N_h	vector with the effective sample sizes used to the left and to the right of the cutoff.
c	cutoff value.
p	order of the polynomial used for estimation of the regression function.
q	order of the polynomial used for estimation of the bias of the regression function.
bws	matrix containing the bandwidths used.
tau_cl	conventional local-polynomial estimate to the left and to the right of the cutoff.
tau_bc	bias-corrected local-polynomial estimate to the left and to the right of the cutoff.
coef	vector containing conventional and bias-corrected local-polynomial RD estimates.
se	vector containing conventional and robust standard errors of the local-polynomial RD estimates.
bias	estimated bias for the local-polynomial RD estimator below and above the cutoff.
beta_Y_p_l	conventional p-order local-polynomial estimates to the left of the cutoff for the outcome variable.
beta_Y_p_r	conventional p-order local-polynomial estimates to the right of the cutoff for the outcome variable.
beta_T_p_l	conventional p-order local-polynomial estimates to the left of the cutoff for the first stage (fuzzy RD).
beta_T_p_r	conventional p-order local-polynomial estimates to the right of the cutoff for the first stage (fuzzy RD).
beta_covs	coefficients of the additional covariates, only returned when covs() are used.
V_cl_l	conventional variance-covariance matrix estimated below the cutoff.
V_cl_r	conventional variance-covariance matrix estimated above the cutoff.
V_rb_l	robust variance-covariance matrix estimated below the cutoff.
V_rb_r	robust variance-covariance matrix estimated above the cutoff.
pv	vector containing the p-values associated with conventional, bias-corrected and robust local-polynomial RD estimates.
ci	matrix containing the confidence intervals associated with conventional, bias-corrected and robust local-polynomial RD estimates.

Author(s)

Sebastian Calonico, University of California, Davis, CA. <scalonico@ucdavis.edu>.
 Matias D. Cattaneo, Princeton University, Princeton, NJ. <cattaneo@princeton.edu>.
 Max H. Farrell, University of California, Santa Barbara, CA. <maxhfarrell@ucsb.edu>.
 Rocio Titiunik, Princeton University, Princeton, NJ. <titiunik@princeton.edu>.

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See Also

[rdbwselect](#), [rdplot](#)

Examples

```
x <- runif(1000,-1,1)
y <- 5 + 3*x + 2*(x>=0) + rnorm(1000)
rdrobust(y,x)
```

rdrobust_RDsenate	<i>RD Senate Data</i>
-------------------	-----------------------

Description

Extract of the dataset constructed by Cattaneo, Frandsen, and Titiunik (2015), which include measures of incumbency advantage in the U.S. Senate for the period 1914-2010.

Usage

```
data(rdrobust_RDsenate)
```

Format

A data frame with 1390 observations on the following 2 variables.

`margin` a numeric vector.

`vote` a numeric vector.

Source

Cattaneo, M. D., Frandsen, B., and R. Titiunik. 2015. [Randomization Inference in the Regression Discontinuity Design: An Application to the Study of Party Advantages in the U.S. Senate](#). *Journal of Causal Inference* 3(1): 1-24.

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Cattaneo, M. D., Frandsen, B., and R. Titiunik. 2015. [Randomization Inference in the Regression Discontinuity Design: An Application to the Study of Party Advantages in the U.S. Senate](#). *Journal of Causal Inference* 3(1): 1-24.

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